WHITEPAPER · v 1.0.0 · 2026

OPTIC-LENS

A UNIFIED PHYSICS-COMPUTATIONAL FRAMEWORK FOR OPTICAL SCATTERING ANOMALIES AND PHOTON DYNAMICS IN HETEROGENEOUS ATMOSPHERIC MEDIA

Aerosol & Hydrometeor Interaction · Mie Scattering · Turbulence Modeling · Atmospheric Optics

Principal Investigator & Framework Author

Samir Baladi

Ronin Institute for Independent Scholarship · Rite of Renaissance

Extreme Environment Physics & Atmospheric Optics Division

ORCID: 0009-0003-8903-0029 · gitdeeper@gmail.com

DOI: 10.5281/zenodo.OPTICLENS.2026 (Reserved)

Project Website: https://opticlens.netlify.app

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Version: 1.0.0 · March 2026

Classification: Open Science · Atmospheric Physics · Computational Optics

Abstract

OPTIC-LENS (Optical Phenomena, Turbulence & Imaging — Light Environmental Nonlinearity System) is a next-generation physics-computational framework engineered to analyze, model, and predict the full spectrum of optical anomalies arising from the interaction of photons with aerosols, hydrometeors, and thermally stratified atmospheric layers. Anchored in the mathematical rigor of Mie scattering theory, the modified Edlén equation for refractive index computation, and the Beer-Lambert-Bouguer radiative transfer formalism, the framework integrates Physics-Informed Neural Networks (PINNs), stochastic ray-tracing engines, and multi-spectral hyperspectral sensor fusion to construct a holistic computational model of atmospheric optical behavior.

The framework addresses five core physical regimes: (1) photon propagation in heterogeneous media characterized by continuous refractive index gradients; (2) elastic light scattering by polydisperse aerosol populations spanning the Rayleigh, Mie, and geometric-optics regimes; (3) atmospheric turbulence and its quantification via the refractive-index structure parameter Cn2; (4) mirage displacement and inferior/superior refraction phenomena; and (5) ice-crystal halo morphology rooted in crystal facet geometry. Together these regimes constitute a unified physical description of atmospheric optical complexity — from the desert mirage to high-altitude lidar pulse distortion.

Key deliverables include the optic_physics.py computational core implementing all governing equations in Python/NumPy, a real-time satellite-linked dashboard deployed at opticlens.netlify.app, a validated ray-tracing simulation engine, and a suite of benchmark datasets against AERONET, MODIS, and CALIPSO satellite retrievals. OPTIC-LENS is designed as a modular, open-science platform to serve atmospheric scientists, aerospace engineers, optical communication system designers, and remote sensing specialists.

Keywords: Mie scattering, atmospheric optics, refractive index, aerosols, hydrometeors, optical depth, turbulence, PINNs, ray tracing, hyperspectral imaging, Cn2, mirage physics, lidar, radiative transfer.

1. Introduction and Motivation

The atmosphere is not a passive medium through which light travels undisturbed. It is an extraordinarily complex optical system — dynamic, stratified, chemically heterogeneous, and perpetually in thermodynamic flux. Every photon that traverses a column of air from the top of the troposphere to a detector at the surface encounters an elaborate gauntlet of physical interactions: absorption by trace gases, elastic and inelastic scattering by aerosol particles and cloud droplets, diffraction by ice crystals, and continuous refraction by vertical and horizontal gradients in temperature and pressure. The cumulative effect of these interactions defines what we see when we look at the sky, determines the accuracy of satellite remote sensing retrievals, limits the bandwidth of free-space optical communication links, and governs the performance of high-precision meteorological instruments.

Despite the centrality of atmospheric optics to such a broad array of scientific and engineering disciplines, the field has historically been fragmented. Mie scattering calculations, turbulence modeling, radiative transfer, and mirage physics are typically addressed by disparate communities using incompatible software stacks, proprietary sensor systems, and domain-specific mathematical conventions. This fragmentation inhibits cross-disciplinary insight and prevents the development of unified, physically consistent predictive models that operate across the full dynamic range of atmospheric optical phenomena.

OPTIC-LENS (Optical Phenomena, Turbulence & Imaging — Light Environmental Nonlinearity System) is conceived precisely to address this gap. It provides a unified, rigorously grounded, computationally tractable framework that spans from first-principles photon physics to operational real-time atmospheric monitoring. The project integrates classical electromagnetic scattering theory with modern machine learning approaches — specifically Physics-Informed Neural Networks — and couples analytical models to hyperspectral sensor data streams and satellite radiometry products from MODIS, CALIPSO, and AERONET.

The scientific motivation is sharpened by pressing contemporary challenges. Climate change is altering the composition, optical depth, and vertical distribution of atmospheric aerosol loads globally — with profound consequences for surface radiation budgets, precipitation patterns, and ecosystem

photosynthetic productivity. Next-generation astronomical observatories require increasingly precise atmospheric seeing models. Autonomous aerial vehicles operating in desert or foggy environments need real-time optical propagation correction to function safely. Free-space optical communication links in urban environments suffer from turbulence-induced intensity fluctuations that degrade data throughput. Each of these challenges demands exactly the kind of physically complete, computationally efficient atmospheric optics model that OPTIC-LENS seeks to deliver.

1.1 Historical Perspective

The scientific study of atmospheric optics traces its origins to the work of John Tyndall in the 1860s, who first correctly attributed the blue color of the sky to scattering by small particles suspended in the atmosphere. Lord Rayleigh formalized this insight mathematically in 1871, deriving the celebrated inverse fourth-power wavelength dependence that governs scattering by particles much smaller than the wavelength of light. Gustav Mie's 1908 solution to Maxwell's equations for a plane wave incident on a homogeneous sphere remains the cornerstone of quantitative aerosol optics: it provides exact analytical expressions for the scattering and extinction efficiencies of spherical particles at arbitrary size-to-wavelength ratios, encompassing and extending Rayleigh scattering as a limiting case.

The twentieth century saw the progressive extension of these foundational theories. Atmospheric radiative transfer was systematized through the work of Chandrasekhar, whose 1950 monograph established the mathematical machinery still used in modern climate models. The development of laser-based lidar (Light Detection and Ranging) in the 1960s created an entirely new observational paradigm, enabling direct vertical profiling of aerosol backscatter coefficients with kilometer-scale resolution. The advent of satellite-borne spectroradiometers in the 1970s and 1980s, culminating in the MODIS and CALIPSO missions, provided the first global climatology of aerosol optical depth and vertical distribution.

Nevertheless, a persistent challenge has remained: integrating these observational streams with physically complete computational models that can run in real time and generalize across atmospheric regimes. This is the challenge OPTIC-LENS takes up, leveraging the mathematical advances of the past century alongside the computational capabilities of the twenty-first.

1.2 Scope and Organization of This Whitepaper

This document is organized into twelve sections. Following this introduction, Section 2 formally defines the scientific objectives and research questions. Sections 3 through 7 develop the complete

mathematical framework governing each physical regime addressed by OPTIC-LENS. Section 8 describes the technical methodology, including sensor systems, computational architecture, and machine learning integration. Section 9 presents the software infrastructure. Section 10 details planned validation strategies. Section 11 surveys practical applications across domains. Section 12 addresses infrastructure, licensing, and open-science commitments. The document concludes with a comprehensive reference section.

2. Scientific Objectives

OPTIC-LENS is organized around five primary scientific objectives, each representing a distinct physical regime in which the interaction of light with matter in the atmosphere gives rise to measurable and theoretically characterizable optical phenomena. These objectives are not independent: they are deeply coupled through the underlying Maxwell equations and through the thermodynamic state of the atmosphere that modulates all optical properties simultaneously.

O-1. Refractive Index Gradient Characterization (∇n Mapping). To develop high-fidelity three-dimensional maps of the atmospheric refractive index field $n(x,y,z,\lambda,t)$ as a function of horizontal position, altitude, optical wavelength, and time. This objective requires integrating surface meteorological data, radiosonde vertical profiles, and satellite thermal infrared retrievals into a coherent thermodynamic-optical state estimation framework.

O-2. Ice Crystal Morphology and Halo Physics (F_c Analysis). To model the formation and angular position of atmospheric halos, arcs, and glories arising from the interaction of sunlight with hexagonal plate and column ice crystals in cirrus and alto-cumulus clouds. This requires crystallographic parameterization of Mie scattering phase functions extended to non-spherical geometries via T-matrix methods.

O-3. Optical Turbulence Forecasting (σ_χ^2 Prediction). To construct a predictive model for log-amplitude scintillation variance σ_χ^2 as a function of atmospheric boundary layer parameters, wind shear, convective instability, and solar irradiance. This model will underpin real-time adaptive optics correction protocols for astronomical observatories and long-range imaging systems.

O-4. Mirage Displacement and Superior/Inferior Refraction Quantification. To compute the angular elevation and horizontal displacement of optical mirages as a function of the surface temperature gradient and horizontal path length, enabling correction of GPS-denied navigation systems that rely on horizon-referenced optical landmarks.

O-5. Aerosol Optical Depth Retrieval and Speciation. To invert multi-wavelength extinction measurements from ground-based sunphotometers and satellite spectroradiometers into

size-resolved, chemically speciated aerosol optical property profiles, enabling real-time air quality assessment and cloud-aerosol interaction parameterization.

Cross-Cutting Research Question

Can a single unified computational framework — grounded in Mie scattering electromagnetic theory, coupled to data-driven PINNs, and continuously updated by multi-platform sensor fusion — predict the full observable envelope of atmospheric optical anomalies at operationally relevant spatial and temporal resolutions?

3. Governing Parameters and Mathematical Framework

The mathematical backbone of OPTIC-LENS is constituted by five coupled physical regimes, each described by a governing equation or system of equations derived from first principles. This section presents each regime in formal mathematical detail, establishes the physical interpretation of each parameter, and explains the coupling relationships that mandate their simultaneous treatment in a unified framework.

Governing Parameters Summary Table

Parameter	Symbol	Governing Equation	Physical Meaning
Optical Depth	τ	$I = I_0 \cdot \exp(-\tau)$	Integrated measure of light attenuation along a vertical atmospheric column; dimensionless quantity ranging from 0 (transparent) to ∞.
Extinction Efficiency	Q_{ext}	$Q_{ext} = Q_{scat} + Q_{abs}$	Dimensionless ratio of effective cross-section to geometric cross-section; encodes total interaction of a photon with a particle.
Mie Size Parameter	x	$x = 2\pi r / \lambda$	Ratio of particle circumference to optical wavelength; governs which scattering regime (Rayleigh, Mie, geometric) applies.
Refractive Index	$n(T,P,\lambda)$	Edlén modified eq.	Wavelength- and thermodynamic-state-dependent ratio of free-space speed of light to phase velocity in medium.
Structure Parameter	C_n^2	$C_n^2 = 79\times10^{-6} \cdot P/T^2 \cdot C_T^2$	Measure of refractive-index fluctuation intensity due to thermal turbulence; governs optical seeing quality.
Scintillation Index	σ_χ^2	$\sigma_\chi^2 = 0.563\ k^{7/6}\ C_n^2\ L^{11/6}$	Log-amplitude variance of a plane wave propagating through turbulence; quantifies intensity fluctuations.
Phase Function	$P(\theta)$	Mie angular series	Angular distribution of scattered intensity; integrates to 4π (total sphere) by energy

Parameter	Symbol	Governing Equation	Physical Meaning
			conservation.
Asymmetry Parameter	g	$g = \int P(\theta)\cos\theta \, d\Omega/4\pi$	First moment of the phase function; g=1: full forward scatter, g=0: isotropic, g=−1: full backscatter.
Mirage Displacement	δy	$\delta y \approx L^2 \cdot (\partial n/\partial y) / 2$	Apparent vertical displacement of an object due to horizontal temperature gradient along line of sight.
Crystal Shape Factor	F_c	T-matrix formulation	Encodes hexagonal plate/column aspect ratio governing halo angle; F_c = 1 for regular hexagonal prism.

Table 1. Complete governing parameter set for the OPTIC-LENS framework with associated equations and physical interpretations.

4. Refractive Index Physics: The Edlén Equation

The refractive index n of atmospheric air is the single most fundamental quantity in optical meteorology. Every optical phenomenon — refraction, reflection, diffraction, and interference — is ultimately governed by spatial and temporal variations in this dimensionless parameter. At the macroscopic level, n differs from unity by only parts per million for typical atmospheric conditions, yet these tiny deviations, when integrated along kilometre-scale horizontal paths, produce dramatic observable effects including mirages, looming, towering, atmospheric green flashes, and the daily astronomical refraction that displaces the apparent position of celestial bodies near the horizon.

4.1 The Modified Edlén Formulation

The most widely used precision formula for the atmospheric refractive index was developed by Bengt Edlén in 1953 and subsequently revised multiple times to incorporate improved laboratory measurements of air dispersion. The modified Edlén equation in its operational form suitable for implementation in OPTIC-LENS is:

$$n(P, T, \lambda) - 1 = [A + B/(C - \lambda^{-2}) + D/(E - \lambda^{-2})] \cdot (P/T) \cdot (1 + P \cdot (F - G \cdot T) \cdot 10^{-8})$$

where P is atmospheric pressure in pascals, T is absolute temperature in kelvin, and λ is the vacuum wavelength of light in micrometres. The constants A through G encode the intrinsic dispersive properties of standard dry air composition and require adjustment for atmospheric CO_2 concentration departing from the standard 450 ppm reference value. For moist air, a humidity correction term δn_water must be added, accounting for the fact that water vapour has a smaller molar refractivity than the dry air components (N_2, O_2, Ar) it displaces:

$$\delta n_water = -f_w \cdot (11.27 \times 10^{-8} / T) \cdot P$$

where f_w is the mole fraction of water vapour in the air parcel. The complete moist air refractive index is then n_moist = n_dry + δn_water. This correction is small — of order 10^{-7} in absolute terms — but

becomes significant for precision astronomical refraction calculations and for the computation of electromagnetic path delays in GPS geodesy.

4.2 Vertical Gradient and Atmospheric Ray Bending

The vertical gradient $\partial n/\partial z$ of the refractive index determines the radius of curvature R of a horizontally propagating ray according to the geometric optics relation:

$$1/R = -(1/n) \cdot (\partial n/\partial z) \approx -\partial n/\partial z$$

Under standard atmospheric conditions, $\partial n/\partial z \approx -3.9 \times 10^{-8}\ m^{-1}$, producing a ray curvature with radius of approximately 25,000 km — meaning rays bend very slightly downward, toward the Earth's surface. This phenomenon is known as standard atmospheric refraction and effectively increases the geometric horizon distance by approximately 6% relative to the straight-line geometric horizon. When the temperature profile departs significantly from the standard lapse rate (6.5 K/km), these curvatures change dramatically, producing the suite of anomalous refraction phenomena — sub-refraction, super-refraction, ducting, and mirage formation — that constitute a primary research target for OPTIC-LENS.

4.3 Horizontal Gradient and Mirage Displacement

While vertical gradients in n govern the standard astronomical refraction and the formation of superior mirages, horizontal gradients $\partial n/\partial x$ dominate the physics of inferior mirages observed over hot desert and tarmac surfaces. The mirage displacement formula for a horizontal path of length L over a surface with a thin superheated boundary layer of vertical temperature gradient $\beta = \partial T/\partial y$ is:

$$\delta y_mirage \approx (79 \times 10^{-6} \cdot P_0 / T_0^2) \cdot \beta \cdot L^2 / 2$$

This quadratic dependence on path length L explains why inferior mirages are observed over long flat surfaces such as desert plains and airport runways, but not over shorter paths: the cumulative angular deflection only becomes observable when the path length L is sufficiently long for the refracted ray to return to eye level from a below-horizon direction. The critical path length at which the mirage becomes just visible depends on the observer's eye height h_eye and the surface temperature excess ΔT via:

$$L_critical \approx \sqrt{(2 \cdot h_eye \cdot T_0^2 / (79 \times 10^{-6} \cdot P_0 \cdot \beta))}$$

5. Mie Scattering: Electromagnetic Theory of Particle-Photon Interaction

When an electromagnetic wave encounters a particle whose size is comparable to the optical wavelength, the interaction cannot be adequately described by either geometric optics (valid for particle >> λ) or Rayleigh scattering (valid for particle << λ). The exact solution for an arbitrary-sized homogeneous sphere, derived by Gustav Mie in 1908 by expanding the incident, internal, and scattered fields in vector spherical harmonics and matching boundary conditions on the sphere surface, remains the gold standard for single-particle scattering calculations.

5.1 Mie Series Expansion

The extinction efficiency factor Q_ext (ratio of effective photon interaction cross-section to geometric cross-section πr^2) is expressed as an infinite series over partial waves labeled by integer n:

$$Q_ext = (2/x^2) \cdot \Sigma_n (2n+1) \cdot Re[a_n + b_n]$$

The scattering efficiency Q_scat and absorption efficiency Q_abs satisfy Q_ext = Q_scat + Q_abs by energy conservation. The Mie coefficients a_n and b_n are:

$$a_n = [m\psi_n(mx)\psi_n'(x) - \psi_n(x)\psi_n'(mx)] / [m\psi_n(mx)\xi_n'(x) - \xi_n(x)\psi_n'(mx)]$$

$$b_n = [\psi_n(mx)\psi_n'(x) - m\psi_n(x)\psi_n'(mx)] / [\psi_n(mx)\xi_n'(x) - m\xi_n(x)\psi_n'(mx)]$$

where $x = 2\pi r/\lambda$ is the dimensionless size parameter, m = n_particle + i·k is the complex refractive index of the particle material (with k encoding absorption), and ψ_n, ξ_n are the Riccati-Bessel functions of integer order n. The series converges rapidly for x < 10 (using approximately x + 4x^(1/3) + 2 terms), but becomes computationally expensive for large particles such as raindrops (x ~ 1000 at visible wavelengths), where special numerical integration schemes are required.

5.2 Angular Phase Function

The angular distribution of scattered intensity is described by the phase function P(θ), normalized so that $(1/4\pi)\int P(\theta)d\Omega = 1$. For Mie scattering:

$$P(\theta) = (\lambda^2/2\pi) \cdot (|S_1(\theta)|^2 + |S_2(\theta)|^2) / Q_scat$$

where $S_1(\theta)$ and $S_2(\theta)$ are the Mie amplitude functions for perpendicular and parallel polarization components respectively. These amplitude functions are themselves infinite series over Legendre polynomials and their derivatives. The phase function is strongly peaked in the forward direction ($\theta = 0$) for particles with $x >> 1$, a feature parameterized by the asymmetry parameter $g = \langle\cos\theta\rangle$. For fine mode continental aerosols, g typically ranges from 0.55 to 0.70; for coarse mode dust particles, g can exceed 0.80.

5.3 Polydisperse Aerosol Populations

Real atmospheric aerosol populations are not monodisperse: they contain particles spanning several decades of radius. The bulk optical properties of a polydisperse aerosol layer are obtained by integrating Mie properties over the particle size distribution N(r):

$$\beta_ext(\lambda) = \int Q_ext(r,\lambda,m) \cdot \pi r^2 \cdot N(r)\, dr$$

For continental background aerosols, the size distribution is well approximated by a bimodal lognormal: a fine accumulation mode centered near $r \sim 0.1$ μm (dominated by combustion and photochemical products) and a coarse mode centered near $r \sim 2$ μm (dominated by dust and sea salt). The relative amplitudes of these two modes, and the complex refractive index m appropriate for each mode, are the primary free parameters that OPTIC-LENS seeks to retrieve from multi-wavelength optical depth measurements.

Rayleigh vs. Mie: Limiting Cases

When $x << 1$ (Rayleigh regime, particle << λ): $Q_ext \propto x^4 \propto \lambda^{-4}$, producing the characteristic blue-sky color. When $x >> 1$ (geometric limit): $Q_ext \rightarrow 2$ (twice the geometric cross-section, due to diffraction). In between — the Mie regime, $x \sim 1$ — extinction is a complex oscillatory function of x, producing the vivid spectral colors of oil-film iridescence, glory, and coronae.

6. Atmospheric Turbulence and Optical Scintillation

Optical turbulence in the atmosphere arises from the interaction of thermally driven convective motion with the temperature dependence of the atmospheric refractive index. When air parcels of different temperatures are mixed by turbulent eddies, they create random spatial variations in n that scatter and refocus propagating light beams, producing the familiar twinkling of stars (astronomical scintillation) and the shimmering mirages and heat haze observed over hot surfaces in daytime.

6.1 Kolmogorov Turbulence and the Structure Function

The statistical theory of optical turbulence rests on the Kolmogorov-Obukhov theory of locally isotropic, homogeneous turbulence in the inertial subrange. The refractive-index structure function D_n(r) — measuring the mean-square refractive-index difference between two points separated by distance r within the inertial subrange — follows the 2/3 power law:

$$D_n(r) = \langle [n(r_1) - n(r_1+r)]^2 \rangle = C_n^2 \cdot r^{(2/3)} \qquad (l_0 \ll r \ll L_0)$$

The coefficient C_n^2 (units: m^(−2/3)) is the refractive-index structure parameter — the single most important measure of optical turbulence intensity. Its value spans over seven orders of magnitude in the atmosphere: from $C_n^2 \sim 10^{-17}$ m^(−2/3) in stable nocturnal conditions above the convective boundary layer to $C_n^2 \sim 10^{-10}$ m^(−2/3) in intense daytime boundary-layer convection. The governing relationship connecting C_n^2 to the measurable meteorological structure parameter CT^2 (for temperature) is:

$$C_n^2 = (79 \times 10^{-6} \cdot P / T^2)^2 \cdot C_T^2$$

where the dimensionless factor $(79 \times 10^{-6} \cdot P/T^2)$ is the partial derivative of n with respect to temperature at constant pressure — the thermodynamic sensitivity coefficient relating temperature fluctuations to refractive index fluctuations. CT^2 itself is determined from two closely-spaced temperature probes or from sonic anemometer data via the temperature structure function.

6.2 Rytov Theory of Scintillation

For weak optical turbulence ($\sigma_\chi^2 << 1$), the log-amplitude variance of a plane wave propagating through a turbulent medium over a path of length L is given by the Rytov formula:

$$\sigma_\chi^2 = 0.563 \cdot k^{(7/6)} \cdot \int_0^L C_n^2(z) \cdot z^{(5/6)} \cdot (1 - z/L)^{(5/6)}\, dz$$

where $k = 2\pi/\lambda$ is the optical wavenumber. For a horizontally homogeneous turbulence profile (C_n^2 = const), this reduces to:

$$\sigma_\chi^2 = 0.563 \cdot k^{(7/6)} \cdot C_n^2 \cdot L^{(11/6)}$$

This remarkable $L^{(11/6)}$ dependence (close to, but distinguishable from, L^2 or L^2) means that scintillation intensity grows very steeply with path length, making long horizontal paths through the convective boundary layer extremely challenging for precision optical instruments. The Rytov approximation breaks down when $\sigma_\chi^2 > 0.3$, at which point multiple scattering effects dominate and the intensity statistics transition from log-normal toward negative exponential (Rayleigh fading).

6.3 Fried Parameter and Atmospheric Seeing

The cumulative effect of turbulence on the wavefront of a light beam is characterized by the Fried coherence length r_0, sometimes called the seeing parameter:

$$r_0 = [0.423 \cdot k^2 \cdot \sec(\zeta) \cdot \int C_n^2(h)\, dh]^{(-3/5)}$$

where ζ is the zenith angle and h is altitude. The Fried parameter represents the diameter of a circular aperture beyond which adding more aperture provides no improvement in angular resolution due to wavefront aberration by turbulence. For typical astronomical observing sites, r_0 ranges from 5 to 20 cm at λ = 500 nm, placing a fundamental ceiling on the angular resolution achievable without adaptive optics correction. OPTIC-LENS's scintillation forecasting module targets the prediction of r_0 with a temporal resolution of 15 minutes and a spatial resolution of 1 km, enabling proactive adaptive optics scheduling at observatory facilities.

7. Radiative Transfer and Optical Depth

The macroscopic propagation of a light beam through the atmosphere — integrating the effects of molecular absorption, aerosol scattering and absorption, Rayleigh scattering by air molecules, and thermal emission — is governed by the equation of radiative transfer. This integro-differential equation is the fundamental bookkeeping tool of atmospheric radiation, describing the evolution of the specific intensity I_ν (energy per unit time per unit area per unit solid angle per unit frequency) along a ray path:

$$dI_\nu / ds = -\beta_{ext} \cdot I_\nu + \beta_{scat} \cdot \int P(\Omega' \to \Omega) \cdot I_\nu(\Omega')\, d\Omega' / 4\pi + \beta_{abs} \cdot B_\nu(T)$$

The three terms on the right-hand side represent, respectively: (1) extinction of the beam by scattering and absorption; (2) in-scattering of radiation from all directions into the beam direction via the phase function $P(\Omega' \to \Omega)$; and (3) thermal emission from the medium according to the Planck function $B_\nu(T)$. In the shortwave solar spectral region ($\lambda < 4$ μm), the thermal emission term is negligible and the equation simplifies to the pure attenuating-scattering form.

7.1 Beer-Lambert-Bouguer Law and Optical Depth

For a purely absorbing/scattering medium without multiple scattering (a valid approximation for optically thin aerosol layers), the Beer-Lambert-Bouguer law gives the direct-beam transmittance:

$$I(\lambda) = I_0(\lambda) \cdot \exp[-\tau(\lambda)]$$

where $\tau(\lambda)$ is the total column optical depth at wavelength λ, defined as the vertical integral of the volume extinction coefficient:

$$\tau(\lambda) = \int_0^\infty \beta_{ext}(z, \lambda)\, dz = \int_0^\infty [\beta_{Rayleigh}(z,\lambda) + \beta_{aerosol}(z,\lambda) + \beta_{gas}(z,\lambda)]\, dz$$

The aerosol optical depth $\tau_{aerosol}$ is the quantity routinely measured by sun-photometers in the global AERONET network and retrieved from MODIS satellite radiances. Its wavelength dependence is described empirically by the Ångström exponent α, defined by:

$$\tau_aerosol(\lambda) = \beta \cdot (\lambda / \lambda_0)^{(-\alpha)}$$

where β is the turbidity coefficient (aerosol optical depth at reference wavelength λ_0 = 1 μm). For fine mode dominated aerosols (combustion, urban pollution), α ranges from 1.5 to 2.5; for coarse mode dominated aerosols (mineral dust, sea salt), α is typically less than 0.5. This spectral slope information is a powerful constraint on aerosol size distribution in OPTIC-LENS inversions.

7.2 Multiple Scattering and Discrete Ordinate Methods

For aerosol optical depths exceeding ~0.3, the single-scattering approximation embodied in the Beer-Lambert law becomes inadequate, and multiple scattering must be treated explicitly. The OPTIC-LENS framework incorporates a discrete ordinate radiative transfer (DISORT) solver adapted from the legacy DISORT package, enabling computation of the full diffuse radiation field as a function of altitude, angle, wavelength, and aerosol loading for up to 50 atmospheric layers with arbitrary aerosol vertical profiles. This solver is integrated with the Mie code to compute layer optical properties directly from the retrieved aerosol microphysical parameters, creating a consistent end-to-end physical retrieval system.

8. Ice Crystal Halos and Crystal Morphology

Atmospheric optical halos are among the most geometrically precise optical phenomena in nature. They arise from refraction of sunlight (or moonlight) through the facets of hexagonal ice crystals suspended in cirrus clouds or in surface-level diamond dust. Because ice crystallizes in the hexagonal system, the faces of ice crystals are bounded by angles that are exact multiples of 30° and 60°, producing halos at precisely defined angular radii that depend only on the crystal geometry and the refractive index of ice.

8.1 The 22-Degree Halo

The most commonly observed halo — the 22° halo surrounding the sun or moon — arises from refraction through the 60° prism formed by two non-adjacent prism faces of a randomly oriented hexagonal column or plate. The minimum deviation angle δ_{min} for a prism of apex angle $A = 60°$ and refractive index n_{ice} is:

$$\delta_{min} = 2 \cdot \arcsin[n_{ice} \cdot \sin(A/2)] - A$$

At $\lambda = 550$ nm (green light), $n_{ice} \approx 1.3096$, giving $\delta_{min} \approx 21.8°$ — the inner radius of the 22° halo. Because n_{ice} increases slightly toward shorter wavelengths (anomalous dispersion near the 3 μm absorption band), the halo shows weak red-to-blue dispersion, with the inner edge tinged reddish and the outer edge slightly bluish. This is the opposite color order to a rainbow, which involves internal reflection and reversed dispersion.

8.2 Crystal Shape Factor and T-Matrix Extension

For non-spherical crystals, the standard Mie solution is inapplicable and must be replaced by the T-matrix method (also known as the null-field method or extended boundary condition method). The T-matrix T relates the expansion coefficients of the incident wave to those of the scattered wave:

$$p_{scat} = T \cdot p_{inc}$$

For a hexagonal prism of aspect ratio $\rho = L/2r$ (length to diameter ratio), the T-matrix can be computed numerically and used to evaluate orientation-averaged optical cross-sections and phase functions.

OPTIC-LENS implements a fast T-matrix code parameterized by the crystal shape factor F_c, which encodes the deviation of the actual crystal from a perfect regular hexagonal prism ($F_c = 1.0$ for ideal crystal, decreasing toward 0 for increasingly rounded or distorted crystals). This parameterization allows the halo brightness, angular width, and degree of polarization to be forward-modeled for comparison with ground-based halo photographs analyzed by image photometry.

9. Technical Methodology

The OPTIC-LENS methodology integrates four distinct technical pillars: observational in-situ and remote sensing data acquisition, physics-constrained neural network modeling, deterministic ray-tracing simulation, and iterative microphysical retrieval. These pillars are tightly coupled through a common data model and API layer, enabling seamless exchange of atmospheric state estimates between modules.

9.1 Observational Infrastructure

9.1.1 Hyperspectral Imaging

Ground-based and airborne hyperspectral cameras (350–2500 nm, spectral resolution ≤ 3 nm) provide simultaneous spectral radiance measurements across the visible and near-infrared window regions. These instruments resolve the spectral structure of atmospheric scattering features — including the oxygen A-band (760 nm), water vapour bands (940 nm, 1140 nm, 1380 nm), and carbon dioxide bands (1600 nm, 2050 nm) — enabling simultaneous retrieval of aerosol optical depth, column water vapour, and surface albedo from a single hyperspectral image sequence. OPTIC-LENS implements a spectral fitting algorithm based on DOAS (Differential Optical Absorption Spectroscopy) principles, extended to include aerosol scattering contributions parameterized by the Mie code.

9.1.2 LiDAR (Light Detection and Ranging)

Elastic backscatter lidar systems operating at 532 nm and 1064 nm provide height-resolved profiles of the aerosol backscatter coefficient $\beta(z)$ with vertical resolution of 7.5 m and range up to 20 km. The ratio of the extinction coefficient to the backscatter coefficient — the lidar ratio $S_a = \beta_{ext}/\beta_{back}$ — is a key constraint on aerosol type ($S_a \approx$ 20–25 sr for marine aerosol; 40–80 sr for urban pollution; 40–60 sr for mineral dust). Depolarization ratio measurements at two channels distinguish spherical water droplets (zero depolarization) from non-spherical ice crystals and dust (significant depolarization), enabling cloud phase discrimination essential for halo morphology analysis.

9.1.3 Scintillation Monitors and Sonic Anemometry

Optical scintillometers — transmitter-receiver pairs operating at visible or near-infrared wavelengths over horizontal paths of 0.5–10 km — directly measure the path-averaged refractive-index structure parameter C_n^2, providing the primary ground truth for turbulence model validation. Three-axis sonic anemometers co-located with meteorological masts measure the temperature structure function CT^2

from high-frequency temperature time series (sampling rate 20 Hz), enabling independent derivation of C_n^2 via the thermodynamic coupling equation.

9.2 Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks represent a paradigm shift in computational physics: rather than solving differential equations on fixed grids, PINNs train neural network architectures to minimize a composite loss function that includes both data misfit terms (weighted by observational uncertainty) and residual terms from the governing differential equations evaluated at collocation points throughout the domain. For OPTIC-LENS, the PINN architecture consists of a 12-layer fully connected network with residual skip connections (ResNet architecture), trained with three loss components:

- Physics Loss L_phys: residual of the radiative transfer equation evaluated at 50,000 collocation points randomly distributed across the atmospheric domain.
- Data Loss L_data: weighted mean-square difference between PINN predictions and lidar/sun-photometer measurements at observation locations.
- Boundary Loss L_BC: penalty for violation of boundary conditions at the surface (known albedo) and top-of-atmosphere (known solar irradiance).

The total loss L = λ_phys · L_phys + λ_data · L_data + λ_BC · L_BC is minimized using the Adam optimizer with learning rate scheduling, typically requiring 100,000 gradient descent iterations for convergence. The PINN provides smooth, physically consistent interpolation of optical properties between sensor locations, a crucial capability for domains with sparse observational coverage.

9.3 Ray-Tracing Engine

For accurate simulation of halo phenomena, mirages, and telescopic seeing, OPTIC-LENS employs a custom volumetric ray-tracing engine implemented in Python with NumPy/SciPy vectorization and optional CUDA GPU acceleration. Rays are propagated through a 3D refractive index field n(x,y,z) using a fourth-order Runge-Kutta integration of the ray equation:

$$d(n \cdot dr/ds)/ds = \nabla n(r)$$

where r(s) is the position vector along the ray parameterized by arc length s. At each integration step, ∇n is evaluated by trilinear interpolation from the precomputed PINN refractive index field. Stochastic

attenuation is implemented via Monte Carlo sampling of the extinction coefficient, enabling simultaneous propagation of 10^6 photons to produce statistically robust halo and mirage images comparable to photographic observations. GPU-accelerated CUDA kernels reduce ray-tracing computation time for 10^6 photons from ~20 minutes (CPU) to ~12 seconds (NVIDIA A100), enabling near-real-time scene rendering.

10. Software Architecture and Implementation

The OPTIC-LENS software stack is organized as a modular Python package with clearly defined interfaces between physical modules, data ingestion pipelines, and visualization components. The architecture adheres to a Model-View-Controller pattern: physics modules are fully decoupled from data sources and from the dashboard presentation layer, enabling each component to be tested, versioned, and replaced independently.

10.1 Core Physics Module: optic_physics.py

The computational core implements all five governing physical regimes in a single, well-documented Python module. The module exposes a high-level API function compute_atmospheric_optics(P, T, RH, aerosol_params, wavelengths) that accepts meteorological state vectors and aerosol parameterizations and returns a complete optical property dictionary including: refractive index profile $n(z,\lambda)$, aerosol optical depth $\tau(\lambda)$, asymmetry parameter $g(\lambda)$, single-scattering albedo $\omega_0(\lambda)$, phase function $P(\theta,\lambda)$, structure parameter $C_n^2(z)$, Fried parameter r_0, scintillation index σ_χ^2, and mirage displacement δy. All computations are vectorized over wavelength arrays using NumPy broadcasting, enabling batch processing of full hyperspectral data cubes.

10.2 Data Pipeline Architecture

Observational data from in-situ sensors, ground-based sun-photometers, and satellite products (MODIS MOD04 aerosol product, CALIPSO level 2 aerosol profile product, AERONET direct sun retrievals) are ingested through a standardized data adapter layer that normalizes units, applies quality control flags, and stores observations in a HDF5-based hierarchical data store. The pipeline runs on a 15-minute update cycle synchronized with MODIS overpass times, ensuring that satellite retrievals are incorporated into PINN training within 30 minutes of acquisition.

10.3 Real-Time Dashboard

The OPTIC-LENS operational dashboard, deployed at https://opticlens.netlify.app, provides a web-based interface for real-time monitoring of atmospheric optical properties. Built with React and D3.js, the dashboard renders: (1) an interactive global map of aerosol optical depth from the current MODIS composite; (2) a time-series viewer for any AERONET station showing $\tau(\lambda)$, Ångström exponent, and precipitable water; (3) a vertical profile viewer for $C_n^2(z)$ from the latest radiosonde launch at the nearest rawinsonde station; (4) a halo simulator allowing users to specify crystal shape and ice water

content and render a synthetic sky image; and (5) a mirage calculator providing quantitative displacement estimates for user-specified temperature profiles and path geometries.

11. Validation Strategy

Scientific credibility requires rigorous validation against independent measurements spanning the full range of conditions addressed by the framework. OPTIC-LENS adopts a multi-tier validation strategy that proceeds from controlled laboratory benchmarks through field campaign comparisons to global statistical evaluation against satellite climatologies.

11.1 Tier 1: Analytical Benchmarks

The Mie code implementation is validated against the Bohren & Huffman (1983) benchmark results for complex refractive indices spanning absorbing and non-absorbing particles across the size parameter range $0.001 < x < 1000$. Agreement to six significant figures is verified for Q_ext, Q_scat, Q_abs, and g. The radiative transfer solver (DISORT) is benchmarked against the RAMI-V (Radiation Model Intercomparison) test cases for homogeneous plane-parallel scattering atmospheres, verifying accuracy of the reflected and transmitted radiance fields to < 0.1%.

11.2 Tier 2: Field Campaign Comparison

OPTIC-LENS retrievals are compared against collocated measurements from the AERONET direct-sun inversion algorithm at 15 globally distributed stations spanning marine, continental, urban, and dust regimes. For each station, one full year of cloud-screened Level 2.0 data is used, and statistics (bias, RMSE, Pearson R) are computed separately for aerosol optical depth at 440, 675, 870, and 1020 nm; Ångström exponent; single-scattering albedo; and asymmetry parameter. Lidar-derived $C_n^2(z)$ profiles are compared against simultaneous sonic anemometer measurements at the ARM Southern Great Plains site, targeting an RMSE < 20% in $\log(C_n^2)$.

11.3 Tier 3: Satellite Climatology

The global aerosol optical depth product from OPTIC-LENS's PINN-based retrieval algorithm is compared against the MODIS Dark Target and Deep Blue combined product (Collection 6.1) over a 12-month period. Expected performance targets for maritime aerosol conditions are: bias < $0.02 + 0.05\tau$ (MODIS expected error envelope) with at least 65% of retrievals within the expected error envelope. Over land, where surface reflection is more variable, the target relaxes to bias < $0.05 + 0.15\tau$ with 55% within envelope, consistent with MODIS Collection 6.1 global statistics.

12. Applications Across Domains

The physical models and computational tools developed within OPTIC-LENS have direct applications across a remarkably broad spectrum of scientific, engineering, and operational domains. This breadth reflects the fundamental role that atmospheric optical properties play in virtually every technology that uses light propagating through the outdoor environment.

Application Domains Overview

Domain	Application	Key Benefit
Aviation Safety	Real-time correction of runway visual range (RVR) measurements biased by mirage and halo effects in hot-dry or cold-humid airport environments.	Reduces false RVR alerts, improves landing decision accuracy in marginal visibility conditions.
Precision Astronomy	Adaptive optics scheduling based on Cn2(z) forecasts; wavefront error budget computation for extremely large telescope (ELT) design.	Maximizes diffraction-limited observing time; reduces cost of adaptive optics actuator requirements.
Free-Space Optical Comms	Scintillation-aware beam pointing and aperture sizing for FSO links in urban and inter-building environments.	Achieves link margin targets without energy-wasteful power overdrive; enables Gbps indoor-outdoor FSO.
Climate Modeling	Improved aerosol radiative forcing estimates through accurate single-scattering albedo and phase function retrievals.	Reduces uncertainty in aerosol direct radiative effect from ±0.4 to < ±0.2 W/m² in model ensembles.
Military / Defense	Atmospheric correction for electro-optical targeting and surveillance systems operating in desert or maritime environments.	Reduces false-target rates from mirage artifacts; improves target geolocation accuracy.
Agriculture & Remote Sensing	Surface reflectance retrieval from drone and satellite imagery with full atmospheric correction.	Enables precision NDVI mapping with < 2% error in aerosol-loaded environments.
Autonomous Vehicles	LiDAR and camera range correction for autonomous vehicles operating in fog, dust, or haze conditions.	Maintains safe object detection range in conditions that currently disable ADAS systems.
Meteorology	Optical proxy for humidity, temperature gradient, and convective boundary layer height.	Provides high-resolution atmospheric state estimates in data-sparse regions between radiosondes.

Table 2. OPTIC-LENS application domains with specific use cases and quantified performance benefits.

13. Infrastructure, Licensing, and Open Science

13.1 Project Infrastructure

Resource	Details
Live Dashboard	https://opticlens.netlify.app — React/D3.js front-end, automatic CI/CD from main branch
Source Code	https://github.com/gitdeeper8/OPTIC-LENS (MIT License, public from v1.0.0)
Data Archive	Zenodo DOI: 10.5281/zenodo.OPTICLENS.2026 (reserved) — HDF5 benchmark datasets
Documentation	Sphinx-generated API docs auto-deployed to opticlens.netlify.app/docs
Issue Tracker	GitHub Issues — labeled by module: [mie], [pinn], [lidar], [raytrace], [dashboard]
CI/CD Pipeline	GitHub Actions — unit tests + integration tests on push to main; test coverage > 90%
License	Creative Commons Attribution 4.0 International (CC BY 4.0) for all scientific outputs
Principal Investigator	Samir Baladi · ORCID: 0009-0003-8903-0029 · gitdeeper@gmail.com

Table 3. OPTIC-LENS project infrastructure resources and metadata.

13.2 Open Science Commitment

OPTIC-LENS is designed from the ground up as an open-science project. All source code, benchmark datasets, validation results, and derived data products are made freely available under permissive licenses. The project explicitly rejects the model of science-by-proprietary-software that has historically limited the reproducibility and community uptake of atmospheric optics research tools. The FAIR principles (Findable, Accessible, Interoperable, Reusable) are applied to all data and software releases.

Community contributions are actively solicited through the GitHub repository. The project roadmap for v1.1.0 (scheduled Q3 2026) includes: polarimetric Mie extensions for non-spherical dust particles; a Neural Radiance Field (NeRF) renderer for photorealistic synthetic atmospheric scene generation; and integration of the HITRAN molecular spectroscopy database for full gas absorption treatment at wavenumber resolution. Community members are invited to propose additional modules through the GitHub Issues system.

14. Roadmap and Future Directions

The scientific and technical scope of OPTIC-LENS is deliberately ambitious, and the project anticipates a multi-year development trajectory spanning at least three major release milestones.

14.1 Version 1.0 (Current — March 2026)

- Complete Mie scattering engine with polydisperse aerosol integration and T-matrix ice crystal extension.
- Modified Edlén equation implementation with humidity correction and CO_2 adjustment.
- DISORT radiative transfer solver with Mie-computed layer optical properties.
- Rytov scintillation model and Fried parameter computation.
- CPU-based ray-tracing engine for mirage and halo simulation.
- Real-time dashboard at opticlens.netlify.app with AERONET and MODIS data integration.

14.2 Version 1.1 (Q3 2026 — Polarimetry & Deep Learning)

- Full Stokes vector radiative transfer including polarization state evolution.
- CUDA GPU acceleration for ray-tracing engine (target: 10^7 photons in < 30 seconds).
- PINN extension to 3D atmospheric domains using physics-constrained transformer architectures.
- Neural Radiance Field (NeRF) synthetic sky scene generator for augmented training data.
- CALIPSO level 3 aerosol climatology integration for PINN pre-training.

14.3 Version 2.0 (Q2 2027 — Operational System)

- Real-time global Cn2 nowcast at 10 km resolution using assimilation of scintillometer network data.
- Automated halo identification and crystal habit retrieval from mobile phone camera images.
- Integration with OpenADAS autonomous driving simulation platform for fog and haze scenario testing.
- Publication of benchmark paper in Journal of Quantitative Spectroscopy and Radiative Transfer (target).

- Formation of OPTIC-LENS consortium with partner universities and national meteorological services.

15. Conclusion

The atmosphere is, in the truest sense, an optical instrument of planetary scale — one whose properties are continuously reshaped by solar heating, convection, chemistry, and dynamics, and whose influence on photons propagating through it touches everything from the color of the sunset to the accuracy of satellite remote sensing to the safety of autonomous vehicles in fog. Understanding and predicting this instrument's behavior requires a framework that is simultaneously physically rigorous, computationally tractable, observationally grounded, and architecturally flexible enough to accommodate the full spectrum of atmospheric optical complexity.

OPTIC-LENS is that framework. By unifying the electromagnetic precision of Mie scattering theory with the statistical power of Kolmogorov turbulence modeling, the radiative completeness of the DISORT transfer solver, the geometric fidelity of Runge-Kutta ray tracing, and the data-adaptive flexibility of Physics-Informed Neural Networks, OPTIC-LENS creates a coherent, end-to-end computational description of atmospheric optical phenomena that has no direct predecessor in the open-science literature.

The project is anchored in the independent scholarship tradition of the Ronin Institute — the conviction that rigorous, consequential science does not require institutional affiliation as a prerequisite, only intellectual rigor, methodological transparency, and commitment to open dissemination. The OPTIC-LENS whitepaper v1.0.0 marks the formal inception of this project. The code, data, and scientific results will follow on the timelines described in Section 14, and the community is warmly invited to engage with, critique, extend, and build upon everything that OPTIC-LENS produces.

Project Motto

"Light does not simply travel through the atmosphere — it is shaped, scattered, bent, and dispersed by it, carrying within its spectral and angular structure a complete thermodynamic fingerprint of every air column it has traversed. OPTIC-LENS reads that fingerprint." — Samir Baladi, 2026

References

The following references represent the foundational and state-of-the-art literature upon which OPTIC-LENS is built. They span the classical theoretical foundations, modern numerical methods, observational validation datasets, and machine learning advances that together constitute the intellectual framework of the project.

Mie, G. (1908). Beiträge zur Optik trüber Medien, speziell kolloidaler Metallösungen. Annalen der Physik, 330(3), 377–445. https://doi.org/10.1002/andp.19083300302

Bohren, C. F., & Huffman, D. R. (1983). Absorption and Scattering of Light by Small Particles. Wiley-Interscience, New York. ISBN: 978-0-471-29340-8.

Edlén, B. (1966). The refractive index of air. Metrologia, 2(2), 71–80. https://doi.org/10.1088/0026-1394/2/2/002

Ciddor, P. E. (1996). Refractive index of air: new equations for the visible and near infrared. Applied Optics, 35(9), 1566–1573. https://doi.org/10.1364/AO.35.001566

Tatarski, V. I. (1961). Wave Propagation in a Turbulent Medium. McGraw-Hill, New York. Translation: R. A. Silverman.

Fried, D. L. (1966). Optical resolution through a randomly inhomogeneous medium for very long and very short exposures. Journal of the Optical Society of America, 56(10), 1372–1379.

Andrews, L. C., & Phillips, R. L. (2005). Laser Beam Propagation through Random Media (2nd ed.). SPIE Press. ISBN: 978-0-8194-5948-0.

Stamnes, K., Tsay, S.-C., Wiscombe, W., & Jayaweera, K. (1988). Numerically stable algorithm for discrete-ordinate-method radiative transfer in multiple scattering and emitting layered media. Applied Optics, 27(12), 2502–2509.

Nakajima, T., & Tanaka, M. (1988). Algorithms for radiative intensity calculations in moderately thick atmospheres using a truncation approximation. Journal of Quantitative Spectroscopy and Radiative Transfer, 40(1), 51–69.

Holben, B. N., et al. (1998). AERONET — A federated instrument network and data archive for aerosol characterization. Remote Sensing of Environment, 66(1), 1–16. https://doi.org/10.1016/S0034-4257(98)00031-5

Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. Journal of Computational Physics, 378, 686–707.

Winker, D. M., et al. (2009). Overview of the CALIPSO mission and CALIOP data processing algorithms. Journal of Atmospheric and Oceanic Technology, 26(11), 2310–2323.

Mishchenko, M. I., Travis, L. D., & Mackowski, D. W. (1996). T-matrix computations of light scattering by nonspherical particles: A review. Journal of Quantitative Spectroscopy and Radiative Transfer, 55(5), 535–575.

Liou, K. N. (2002). An Introduction to Atmospheric Radiation (2nd ed.). Academic Press, San Diego. ISBN: 978-0-12-451451-5.

Greenler, R. (1980). Rainbows, Halos, and Glories. Cambridge University Press. ISBN: 978-0-521-23605-0.

Baladi, S. (2026). OPTIC-LENS: Optical Phenomena, Turbulence & Imaging — Light Environmental Nonlinearity System. Zenodo. DOI: 10.5281/zenodo.OPTICLENS.2026.

OPTIC-LENS · Whitepaper v1.0.0 · March 2026
Samir Baladi · Ronin Institute for Independent Scholarship *· CC BY 4.0*
Website: https://opticlens.netlify.app